DQH-BAN: DUAL-STREAM QUANTUM HEALTHCARE INTELLIGENCE (DQH-BAN-EXP-2026-V2)

STREAM A: Hybrid Quantum-Classical Ensemble

Network Flows -> Periodic Coord + Soft Quantile Bins
+ Multi-Scale 1D-CNN (SE) -> Causal TCN -> BiGRU -> Attention
-> 4-Qubit PennyLane VQC + XGB + LGB + RF + LogReg
-> 3-Fold OOF Stacking (Regularized Meta-LR)
-> Validation-Calibrated Threshold (FPR <= 1% / Balanced F1)

STREAM B: Biomedical Intelligence

Physiological Signals (125 Hz ECG, PPG, Resp)
-> Bandpass Filtering + HRV/PTT/SQI Extraction
-> Multi-Scale 1D-CNN + Modality-Masked Gated Fusion
-> Patient-Independent AF / Non-AF Classification
(0% Subject Overlap Across Train / Val / Test)

HYPERLEDGER FABRIC & SECURITY ORCHESTRATION

Device Auth & Access Control | AES-256-GCM Encrypted Off-Chain Vault | SHA-256 Integrity Commitments